

FUAD AHMED (TUBA, FUAD AHMED)

STATEMENT OF PURPOSE & MOTIVATION LETTER

Program: Erasmus Mundus Joint Master in Green Networking & Cloud Computing (GENIAL)

Consortium: Université de Lorraine (France) • Leeds Beckett University (UK) • Luleå University of Technology (Sweden)

Applicant: fuadahmedt@gmail.com • +251 925 278 350 • Addis Ababa, Ethiopia

To the Academic Selection Committee of the Erasmus Mundus GENIAL Consortium,

I am writing to express my profound motivation to pursue the **Erasmus Mundus Joint Master Degree in Green Networking and Cloud Computing (GENIAL)**. As a computer science graduate with four years of professional software systems engineering experience, I have witnessed first-hand the explosive computational demands of distributed architectures and real-time data pipelines. Yet, while hyperscale industry paradigms celebrate latency reductions and processing scale, humanity faces an inescapable physical constraint: the soaring carbon footprint and energy consumption of global digital infrastructure. My ultimate academic objective is to engineer energy-conscious, resource-efficient distributed networks and adaptive cloud-edge scheduling frameworks that harmonize computational power with environmental sustainability.

1. ORIGIN OF MY CONVICTION: COMPUTING UNDER REAL-WORLD RESOURCE CONSTRAINTS

My perspective on computational efficiency was forged in an environment where energy availability and bandwidth limitations are visceral operational realities rather than theoretical trade-offs. Growing up and working as an engineer in Ethiopia, intermittent power stability and network degradation taught me that efficiency is the primary prerequisite of reliable software. When building backend microservices, wasteful algorithmic complexity or unoptimized polling mechanisms do not merely increase electricity bills—they precipitate cascading failures across resource-constrained edge devices. This reality instilled in me a disciplined engineering philosophy: every CPU cycle, memory allocation, and transmitted byte carries a quantifiable physical cost.

2. ACADEMIC FOUNDATIONS & TECHNICAL TRAJECTORY

During my Bachelor of Science in Computer Science at St. Mary's University (GPA: 3.20/4.00, 100% English Medium of Instruction), I built a rigorous mathematical and algorithmic foundation across Operating Systems, Distributed Systems, Computer Networks, and Architecture. For my undergraduate capstone project, I designed and evaluated a high-concurrency, fault-tolerant distributed data synchronization protocol. By implementing lightweight state serialization and dynamic delta-updates, I minimized redundant network packet transmission and CPU thread contention across multi-node topologies.

Over the subsequent four years, I translated these academic principles into scalable enterprise production systems. Working with Python (AsyncIO), Go, and Docker on Linux environments, I architected distributed caching topologies utilizing Redis clusters that curtailed primary database I/O by 48%, significantly diminishing server-side thermal dissipation and energy draw during peak traffic periods. However, optimizing application-layer microservices revealed to me that true sustainability requires holistic, cross-layer co-design—unifying low-level network protocol efficiency, hardware telemetry, dynamic workload placement, and green energy availability.

3. WHY THE GENIAL CONSORTIUM TRIAD IS MY EXACT ACADEMIC HOME

The GENIAL curriculum presents an unparalleled international consortium precisely tailored to my research ambitions:

- **Semester 1 (Université de Lorraine, France):** Solidifying foundational competencies in green networking, eco-design methodologies, and energy-aware protocol verification at the esteemed CRAN research center. France's

leadership in low-carbon energy infrastructure makes it the ideal environment to master network power-consumption modeling.

- **Semester 2 (Leeds Beckett University, United Kingdom):** Deepening my expertise in sustainable software engineering, green data science, and intelligent IoT systems, understanding how software architecture directly dictates hardware energy profiles.
- **Semester 3 (Luleå University of Technology, Sweden):** Specializing in resource-efficient distributed cloud systems, edge computing, and extreme-scale data center energy management. Sweden's northern data center hubs (utilizing passive arctic cooling and 100% renewable hydroelectric power) represent the frontier of global sustainable computing.

4. CAREER VISION & VALUE TO THE ERASMUS COMMUNITY

Following completion of the GENIAL Master's degree, I intend to lead research at the vanguard of carbon-aware distributed computing, either progressing toward doctoral studies or engineering next-generation green cloud orchestrators for industrial research consortia. Long term, I aspire to bridge European technological breakthroughs with emerging digital ecosystems in Africa, pioneering decentralized edge infrastructures powered by renewable microgrids.

I bring to the GENIAL cohort relentless technical dedication, resilience, and a distinctive operational perspective shaped by African technological infrastructure. Supported by the Erasmus Mundus Joint Scholarship, I am prepared to dedicate my full intellectual capacity to the academic excellence and global mission of the GENIAL consortium.

Respectfully submitted,

Fuad Ahmed (TUBA, FUAD AHMED)

BSc in Computer Science, St. Mary's University
Addis Ababa, Ethiopia • fuadahmedt@gmail.com